Q1. What is data science?

A. Data science is the field of extracting insights, knowledge, and value from data using statistics, programming, and machine learning.

Q2. Supervised learning?

A. Supervised learning trains models using labeled data to predict outcomes or classify data.

Q3. Unsupervised learning?

A. Unsupervised learning discovers patterns and relationships in unlabeled data.

Q4. Reinforcement learning?

A. Reinforcement learning trains agents through rewards and penalties based on actions in an environment.

Q5. Classification vs regression?

A. Classification predicts categories, while regression predicts continuous numerical values.

Q6. What is overfitting?

A. Overfitting occurs when a model learns training data too closely and performs poorly on unseen data.

Q7. What is underfitting?

A. Underfitting occurs when a model is too simple to capture underlying patterns in the data.

Q8. Cross-validation?

A. Cross-validation evaluates model performance by splitting data into multiple training and validation sets.

Q9. Feature engineering?

A. Feature engineering creates or transforms variables to improve model performance.

Q10. Feature selection?

A. Feature selection identifies the most relevant variables for model training.

Q11. What is PCA?

A. Principal Component Analysis (PCA) reduces dimensionality while preserving most data variance.

Q12. Dimensionality reduction?

A. Dimensionality reduction decreases the number of features while retaining important information.

Q13. Bias-variance tradeoff?

A. The bias-variance tradeoff balances model simplicity and complexity to improve generalization.

Q14. Confusion matrix?

A. A confusion matrix summarizes classification results using true positives, true negatives, false positives, and false negatives.

Q15. Precision vs recall?

A. Precision measures prediction accuracy for positive cases, while recall measures the ability to find all positive cases.

Q16. F1 score?

A. The F1 score is the harmonic mean of precision and recall.

Q17. ROC-AUC?

A. ROC-AUC measures a model's ability to distinguish between classes across different thresholds.

Q18. Logistic regression?

A. Logistic regression is a classification algorithm that predicts probabilities using a logistic function.

Q19. Linear regression?

A. Linear regression models relationships between variables to predict continuous values.

Q20. Gradient descent?

A. Gradient descent is an optimization algorithm used to minimize model error.

Q21. Random forests?

A. Random Forest is an ensemble learning method that combines multiple decision trees.

Q22. Decision trees?

A. Decision trees split data into branches based on feature values to make predictions.

Q23. What is XGBoost?

A. XGBoost is a high-performance gradient boosting algorithm known for accuracy and efficiency.

Q24. What is clustering?

A. Clustering groups similar data points without predefined labels.

Q25. K-Means?

A. K-Means is a clustering algorithm that partitions data into K groups based on similarity.

Q26. DBSCAN?

A. DBSCAN is a density-based clustering algorithm that identifies clusters and outliers.

Q27. What is NLP?

A. Natural Language Processing enables computers to understand, analyze, and generate human language.

Q28. Tokenization?

A. Tokenization splits text into smaller units such as words, subwords, or sentences.

Q29. Embeddings?

A. Embeddings are numerical vector representations that capture semantic meaning.

Q30. Recommendation systems?

A. Recommendation systems suggest relevant products or content based on user behavior and preferences.

Q31. A/B testing?

A. A/B testing compares two versions of a product or feature to determine which performs better.

Q32. Hypothesis testing?

A. Hypothesis testing uses statistical methods to evaluate assumptions about data.

Q33. What is a p-value?

A. A p-value measures the probability of obtaining results assuming the null hypothesis is true.

Q34. Normal distribution?

A. A normal distribution is a symmetric bell-shaped probability distribution.

Q35. Data cleaning?

A. Data cleaning identifies and corrects errors, inconsistencies, and inaccuracies in datasets.

Q36. Handling missing values?

A. Missing values can be removed, imputed, or estimated using statistical techniques.

Q37. Data leakage?

A. Data leakage occurs when information unavailable in production influences model training.

Q38. What is ETL?

A. ETL stands for Extract, Transform, and Load, a process for moving and preparing data.

Q39. SQL vs Pandas?

A. SQL is used for querying databases, while Pandas is a Python library for data analysis and manipulation.

Q40. What is Apache Spark?

A. Apache Spark is a distributed computing framework for large-scale data processing.

Q41. What is Hadoop?

A. Hadoop is an ecosystem for distributed storage and processing of big data.

Q42. Data pipelines?

A. Data pipelines automate the movement and transformation of data between systems.

Q43. Model deployment?

A. Model deployment makes trained machine learning models available for real-world use.

Q44. Model drift?

A. Model drift occurs when data patterns change, reducing model performance over time.

Q45. Evaluation metrics?

A. Common metrics include accuracy, precision, recall, F1-score, ROC-AUC, RMSE, and MAE.

Q46. Describe a DS project

A. Explain the business problem, data collection, preprocessing, modeling, evaluation, deployment, and results.

Q47. Failure and lesson learned

A. Describe a project challenge, the root cause, corrective actions, and key lessons learned.

Q48. Explain results to stakeholders

A. Communicate findings in simple business terms using visualizations and actionable insights.

Q49. How do you choose features?

A. Features are selected based on domain knowledge, statistical analysis, importance scores, and business relevance.

Q50. Questions for us?

A. Ask about team structure, datasets, tools, project goals, growth opportunities, and success metrics.